



STRUCTURED NOTES

CAPTURE ORGANIZE UNDERSTAND APPLY

Name Andrea Pebles
Course / Subject P.M.
Speaker / Class Carlos Pichardo
Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

NOTES

Title: Logica matematica
Topic: Argumentos validos y no validos

Un argumento esta formado por premisas y una conclusion. Es valido cuando la conclusion se deriva logicamente de las premisas; es decir, no puede ocurrir que todas las premisas sean verdaderas y la conclusion sea falsa

Focus on the key concepts that generate value and understanding

ACTIONS / NEXT STEPS

Turn knowledge into action.
Small steps, big results.

QUESTIONS & REFLECTIONS

What questions do I have?

What is unclear or confusing?

What ideas connect?

How can I apply this?

What assumptions should I challenge?

SUMMARY

Write a brief summary of the main points



KEY TAKEAWAY

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KEYWORDS & KEY CONCEPTS

NOTES

Title: Logica MatematicaTopic: Equivalencia logica

Dos proposiciones son logicamente equivalentes cuando tienen el mismo valor de verdad en todas las combinaciones posibles. La equivalencia puede comprobarse mediante tablas de verdad o utilizando leyes logicas para transformar una expresion en otra sin cambiar su significado

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QUESTIONS & REFLECTIONS

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Name Andrea Febles
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KEYWORDS & KEY CONCEPTS

NOTES

Title: Logica Matemática
Topic: Inferencia logica

La inferencia logica consiste en obtener una conclusion a partir de una o varias premisas mediante reglas de razonamiento. Entre las reglas importantes se encuentran el modus tollens y la transitividad, que permiten determinar conclusiones a partir de informacion conocida

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KEYWORDS & KEY CONCEPTS

NOTES

Title: Logica matematica
Topic: Contingencia

Una contingencia es una proposicion compuesta que resulta verdadera en algunas combinaciones y falsa en otras. Su valor depende de los valores de verdad de las proposiciones que la forman

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Name Andrea Febles

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Speaker / Class Carlos Richarcho

Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

NOTES

Title: Logica Matematica

Topic: Contradiccion

- Una contradiccion compuesta que resulta falsa en todas las combinaciones posibles de valores de verdad. Por ejemplo, es una contradiccion porque una proposicion no puede ser verdadera y falsa al mismo time.

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KEYWORDS & KEY CONCEPTS

- Tautologia
- contradiccion
- contingencia
-
-

Focus on the key concepts that generate value and understanding

ACTIONS / NEXT STEPS

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QUESTIONS & REFLECTIONS

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Name Andrea Pebles
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Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

• Tabla de verdad
• combinaciones
•
•
•

 Focus on the key concepts that generate value and understanding

NOTES

Title: Logica Matematica
Topic: Tablas de verdad

Las tablas de verdad muestran el resultado de una proposición lógica para todas las combinaciones posibles de valores verdaderos y falsos de sus proposiciones simples. Si una expresión contiene n proposiciones diferentes, su tabla tiene 2^n filas.

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QUESTIONS & REFLECTIONS

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SUMMARY

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Las tablas permiten ilustrar el comportamiento de una lógica

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Date & Time: 9-10-2026

Name: Andrea Febles
Course / Subject: P.M.
Speaker / Class: Carlos Pichardo
Date & Time: 9-10-2026

KEYWORDS & KEY CONCEPTS

NOTES

Title: Logica Matemática
Topic: Proposición bicondicional ($\leftrightarrow$)

- Bicondicional
- equivalencia
- condicion
-
-
-

Focus on the key concepts that generate value and understanding

La proposición bicondicional se representa como $p \leftrightarrow q$ y se lee "p si y solo si q". Es verdadera cuando proposiciones tienen el mismo valor de verdad y falsa cuando sus valores son diferentes.

ACTIONS / NEXT STEPS

Turn knowledge into action.
Small steps, big results

QUESTIONS & REFLECTIONS

¿Cuándo es verdadera una condición bicondicional?

What questions do I have?

What is unclear or confusing?

What ideas connect?

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Name Andrea Febles
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Date & Time 4-10-2026

KEYWORDS & KEY CONCEPTS

- condicional
- antecedente
- consecuente
-
-
-



Focus on the key concepts that generate value and understanding

NOTES

Title: Logica matematica
Topic: Proposicion condicional ($\rightarrow$)

La proposicion condicional establece una relacion entre una condicion y un resultado. Se representa como $p \rightarrow q$ y se lee "si p, entonces q". Es falsa cuando p es verdadera y q es falsa; en los demas casos es verdadera.

ACTIONS / NEXT STEPS

Turn knowledge into action.
Small steps, big results.

QUESTIONS & REFLECTIONS

¿En que caso una condicional es falsa?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points

Las condicionales permiten presentar reglas y condiciones utilizadas en prog.



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Name Andrea Febles
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KEYWORDS & KEY CONCEPTS

- conjuncion
- disyuncion
- negacion
-
-
-

Focus on the key concepts that generate value and understanding

ACTIONS / NEXT STEPS

Turn knowledge into action.
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NOTES

Title: Logica matematica
Topic: Proposiciones compuestas

Una proposicion compuesta se forma al conectar dos o mas proposiciones simples mediante operadores logicos. Entre los principales estan la conjuncion (y), la disyuncion(o) y la negacion(no), que permiten construir enunciados con diferentes condiciones de verdad

QUESTIONS & REFLECTIONS

¿Como se forma una proposicion compuesta?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points

Los operadores logicos permiten expresar relaciones entre varias afirmaciones



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